Stat 145 (Multivariate Statistics)

Lesson 2.4 Multivariate Analysis of Variance

Learning Outcomes

1. Derive the statistics for testing hypotheses on $k \geq 3$ mean vectors.
2. Test hypotheses on $k \geq 3$ mean vectors.

Introduction

Often, more than two populations need to be compared. Random samples, collected from each of k populations, are arranged as

	Sample 1 from $N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$	Sample 2 from $N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$	$\cdots$	Sample k from $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
	$\mathbf{y}_{11}$	$\mathbf{y}_{21}$	$\cdots$	$\mathbf{y}_{k1}$
	$\mathbf{y}_{12}$	$\mathbf{y}_{22}$	$\cdots$	$\mathbf{y}_{k2}$
	$\vdots$	$\vdots$		$\vdots$
	$\mathbf{y}_{1n}$	$\mathbf{y}_{2n}$	$\cdots$	$\mathbf{y}_{kn}$
Total	$\mathbf{y}_{1.}$	$\mathbf{y}_{2.}$	$\cdots$	$\mathbf{y}_{k.}$
Mean	$\bar{\mathbf{y}}_1$	$\bar{\mathbf{y}}_2$	$\cdots$	$\bar{\mathbf{y}}_k$
Variance	$\boldsymbol{S}_1$	$\boldsymbol{S}_2$	$\cdots$	$\boldsymbol{S}_k$

where $y_{i.} = \sum_{j=1}^{n} y_{ij}$ and $\bar{y}_{i.} = \frac{\sum_{j=1}^{n} y_{ij}}{n}$. The k samples are assumed to be independent.

In the multivariate case, we assume that k independent random samples of size n are obtained from p-variate normal populations with equal covariance matrices.

The statistical model is

- **Effects model**: $y_{ij} = \mu + \tau_i + \epsilon_{ij},\ i = 1, 2, \cdots, k;\ \ j = 1, 2, \cdots, n$

- **Means model**: $y_{ij} = \mu_i + \epsilon_{ij}, \; i = 1, 2, \cdots, k; \;\; j = 1, 2, \cdots, n$

Why run a Multivariate ANOVA (MANOVA) instead of series of univariate ANOVA for each variable? The univariate ANOVAs, run separately, cannot take into account the pattern of covariance among the variables. Thus, it is possible that even though there may be an experimental effect, no single ANOVA will detect a significant difference.

We wish to compare the mean vectors of the k samples for significant differences. Hence, the null hypothesis is

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

and this will be tested against the alternative

$$H_1 : \text{at least two } \mu' s \text{ are unequal}$$

Equality of the mean vectors implies that the k means are equal for each variable.

$$H_0 : \begin{pmatrix} \mu_{11} \\ \mu_{12} \\ \vdots \\ \mu_{1p} \end{pmatrix} = \begin{pmatrix} \mu_{21} \\ \mu_{22} \\ \vdots \\ \mu_{2p} \end{pmatrix} = \cdots = \begin{pmatrix} \mu_{k1} \\ \mu_{k2} \\ \vdots \\ \mu_{kp} \end{pmatrix}$$

Thus, H_0 implies p sets of equalities.

$$\begin{aligned} \mu_{11} &= \mu_{21} = \cdots = \mu_{k1} \\ \mu_{12} &= \mu_{22} = \cdots = \mu_{k2} \\ &\vdots \\ \mu_{1p} &= \mu_{2p} = \cdots = \mu_{kp} \end{aligned}$$

The *hypothesis* matrix $\mathbf{H}_{p\times p}$ is defined as

$$\begin{aligned} \mathbf{H} &= n \sum_{i=1}^{n} (\bar{y}_{i.} - \bar{y}_{..})(\bar{y}_{i.} - \bar{y}_{..})' \\ &= \sum_{i=1}^{n} \frac{1}{n} y_{i.} y_{i.}' - \frac{1}{kn} y_{..} y_{..}' \\ &= \begin{bmatrix} SSH_{11} & SPH_{12} & \dots & SPH_{1p} \\ SPH_{12} & SSH_{22} & \dots & SPH_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ SPH_{1p} & SPH_{2p} & \dots & SSH_{pp} \end{bmatrix} \end{aligned}$$

where

- $SSH_{rr} = n \sum_{i=1}^{k} (\bar{y}_{i.r} - \bar{y}_{..r})^2 \implies$ Between sum of squares for each variable

- $SPH_{rs} = n\sum_{i=1}^{k}(\bar{y}_{i.r} - \bar{y}_{..r})(\bar{y}_{i.s} - \bar{y}_{..s}) \implies$ Between sum of products for each pair of variables

The degrees of freedom for the **H** matrix is $\nu_H = k - 1$.

The *error* matrix $\mathbf{E}_{p\times p}$ is defined as

$$\begin{aligned}
\mathbf{E} &= \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})(y_{ij} - \bar{y}_{i.})' \\
&= \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}y_{ij}' - \sum_{i=1}^{k}\frac{1}{n}y_{i.}y_{i.}' \\
&= \begin{bmatrix} SSE_{11} & SPE_{12} & \dots & SPE_{1p} \\ SPE_{12} & SSE_{22} & \dots & SPE_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ SPE_{1p} & SPE_{2p} & \dots & SSE_{pp} \end{bmatrix}
\end{aligned}$$

where

- $SSE_{rr} = \sum_{i=1}^{k}\sum_{i=1}^{n}(\bar{y}_{ijr} - \bar{y}_{i.r})^2 \implies$ Within sum of squares for each variable

- $SPE_{rs} = n\sum_{i=1}^{k}\sum_{i=1}^{n}(\bar{y}_{ijr} - \bar{y}_{i.r})(\bar{y}_{ijs} - \bar{y}_{i.s}) \implies$ Within sum of products for each pair of variables

The degrees of freedom for the **E** matrix is $\nu_E = k(n-1)$. Note that

$$E\left(\frac{\mathbf{E}}{k(n-1)}\right) = \Sigma$$

Test statistics for MANOVA

1. Wilk's Lambda Statistic

The Wilk's lambda statistic is given by

$$\Lambda = \frac{|\mathbf{E}|}{|\mathbf{H}+\mathbf{E}|} \implies \frac{\text{within generalized variance}}{\text{total generalized variance}},\ 0 \le \Lambda \le 1$$

We reject H_0 if Λ is too small, that is when $\Lambda \le \Lambda_{\alpha,p,\nu_H,\nu_E}$, where $\nu_H = k-1$ is the degrees of freedom for the hypothesis, and $\nu_E = k(n-1)$ is degrees of freedom for error.

Properties of Wilks' Lambda

1. We need $v_E = k(n-1) \geq p$ for the determinants to be positive.
2. The degrees of freedom for error and hypothesis are the same as for univariate ANOVA.
3. The distribution of Λ_{p,ν_H,ν_E} is the same as Λ_{p,ν_E,ν_H} . This saves some space for the table of critical values.
4. Wilks' Λ can be written as

$$\Lambda = \prod_{i=1}^{s} \frac{1}{1+\lambda_i}$$

where λ_i is the i^{th} eigenvalue of $\mathbf{E}^{-1}\mathbf{H}$. Here $s = min(p; \nu_H)$ is the rank of $\mathbf{H}$, which is also the number of nonzero eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$.

4. Λ is in the interval $[0,1]$. If the sample mean vectors were all equal (for example, if they were all equal to their expected values under the null), then $\mathbf{H} = \mathbf{0}$. This means that when the null hypothesis is true then, $\mathbf{H} \to \mathbf{0}$ and $\Lambda \to 1$, Meanwhile, when the μ_i's are different, then $\Lambda \to 0$.
5. Increasing the number of variables p decreases the critical value for Λ needed to reject the null hypothesis. This means that it is more difficult to reject H_0 (since we reject for small Λ) unless the null hypothesis is false for the new variables. That is, adding new variables for which the populations are equal makes it harder to reject the null hypothesis.
6. When $\nu_H = 1, 2$ or $p = 1, 2$, Wilks' Λ is equivalent to an F statistic. Otherwise, an approximate transformation to an F can be used.

Parameters p, ν_H	Statistic Having F-Distribution	Degrees of Freedom
Any p, $\nu_H = 1$	$\frac{1-\Lambda}{\Lambda} \frac{\nu_E - p + 1}{p}$	p, $\nu_E - p + 1$
Any p, $\nu_H = 2$	$\frac{1-\sqrt{\Lambda}}{\sqrt{\Lambda}} \frac{\nu_E - p + 1}{p}$	$2p$, $2(\nu_E - p + 1)$
$p = 1$, any ν_H	$\frac{1-\Lambda}{\Lambda} \frac{\nu_E}{\nu_H}$	ν_H, ν_E
$p = 2$, any ν_H	$\frac{1-\sqrt{\Lambda}}{\sqrt{\Lambda}} \frac{\nu_E - 1}{\nu_H}$	$2\nu_H$, $2(\nu_E - 1)$

For other values of p and ν_H, an approximate F statistic is given by

$$F = \frac{1-\Lambda^{1/t}}{\Lambda^{1/t}} \left(\frac{df_2}{df_1} \right)$$

where:

$$\begin{aligned}
df_1 &= p\nu_H \\
df_2 &= wt - \frac{1}{2}(p\nu_H - 2) \\
w &= \nu_E + \nu_H - \frac{1}{2}(p + \nu_H + 1) \\
t &= \sqrt{\frac{p^2\nu_H^2 - 4}{p^2 + \nu_H^2 - 5}}
\end{aligned}$$

When $p\nu_H = 2$, $t \stackrel{set}{=} 1$. The approximate F statistic reduces to the exact F statistics in the above table when either p or ν_H is 1 or 2.

If the null hypothesis is rejected, then follow up tests could be made. Fixing $r \in \{1, 2, \cdots, p\}$, one could test

$$H_0 : \mu_{1r} = \mu_{2r} = \cdots = \mu_{kr}$$

which would be a univariate ANOVA test to see if the k populations differ on variable r.

As usual, testing all variables simultaneously and then testing individual variables has better type I error than just testing all variables separately to begin with.

It is also possible that the simultaneous test rejects H_0 but that each H_{0r} for $r = 1, 2, \cdots, p$ fails to be rejected.

2. Roy's Largest Root Statistic

The Roy's largest root statistic is

$$\theta = \frac{\lambda_1}{1 + \lambda_1}$$

where λ_1 is the largest eigenvalue of $\mathbf{E}^{-1}\mathbf{H}$. This means that Roy's largest root statistic uses the variance from the variable that separates the group most based on the largest eigenvalue.

The null hypothesis is rejected if $\theta \geq \theta_{\alpha,s,m,N}$, where $s = min(p, \nu_H)$, $m = \frac{1}{2}(|\nu_H - p| - 1)$, and $N = \frac{1}{2}(\nu_H - p - 1)$.

3. Pillai's Trace Statistic

The Pillai statistic is given by

$$V^{(s)} = tr[(\mathbf{E} + \mathbf{H})^{-1}\mathbf{H}] = \sum_{i=1}^{s} \frac{\lambda_i}{1 + \lambda_1}$$

Looking at the form of the Pillai's V statistic, it is clear that it is an extension of the Roy's largest root statistic. We reject H_0 if $V^{(S)} \geq V_\alpha^{(S)}$. This statistic is considered more conservative than Wilk's lambda.

4. Lawley-Hotelling's Statistic

The Lawley-Hotelling's statistic is given by

$$U^{(s)} = tr[\mathbf{E}^{-1}\mathbf{H}] = \sum_{i=1}^{s} \lambda_i$$

The null hypothesis is rejected in favor of the alternative if $\frac{\nu_E}{\mu_H} U^{(S)}$. This test statistic is considered more liberal than Wilk's Λ.

Assumptions of MANOVA

1. Multivariate normality
2. Homogeneous Variance-Covariance Matrices
3. Independent Observations
4. Response variables are not strongly correlated (multicollinearity)

<u>Example 2.4.1</u>

In a classic experiment carried out from 1918 to 1934, growth of apple trees of six different rootstocks were compared on four measures of size. How do the measures of size vary with the type of rootstock?

The data for eight trees (n=8) from each of six rootstocks (k=6) are collected. The variables are:

- y1 = trunk girth at 4 years (mm x 100)
- y2 = extension growth at 4 years (m)
- y3 = trunk girth at 15 years (mm x 100)
- y4 = weight of tree above ground at 15 years (lb x 1,000)

We would like to test

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

versus

$$H_1 : \mu_i \neq \mu_j, \text{ for at least one pair of } i \neq j$$

```r
library(readxl)
library(tidyverse)
RS <- read_xlsx("RootStock.xlsx")

RS1 <- RS %>%
  rename(girth4 = y1, ext_growth = y2, girth15 = y3, weight15 = y4) %>%
  mutate(Stock = factor(rootstock))
head(RS1)
```

```
## # A tibble: 6 x 6
##   rootstock girth4 ext_growth girth15 weight15 Stock
##       <dbl>  <dbl>      <dbl>   <dbl>    <dbl> <fct>
## 1         1   1.11       2.57    3.58    0.76  1
## 2         1   1.19       2.93    3.75    0.821 1
## 3         1   1.09       2.86    3.93    0.928 1
## 4         1   1.25       3.84    3.94    1.01  1
## 5         1   1.11       3.03    3.6     0.766 1
## 6         1   1.08       2.34    3.51    0.726 1
```

The following code chunk produces the mean vectors for each of the Rootstocks.

```r
k <- 6 # number of groups
p <- 4 # number of variables
n <- 8 # number of observations per group

MVBG <- matrix(NA, k, p) # mean vectors by group
#each row is a mean vector of each Rootstock

for (i in 1:k) {
MVBG[i,] <- round(colMeans(RS[RS$rootstock==i,-1]),4)
}
MVBG
```

```
##        [,1]   [,2]   [,3]   [,4]
## [1,] 1.1375 2.9771 3.7388 0.8711
## [2,] 1.1575 3.1091 4.5150 1.2805
## [3,] 1.1075 2.8152 4.4550 1.3914
## [4,] 1.0975 2.8798 3.9062 1.0390
## [5,] 1.0800 2.5573 4.3125 1.1810
## [6,] 1.0363 2.2146 3.5962 0.7350
```

To generate the **H** and **E** matrices we use the following code chunks.

```r
H <- round(n*(k-1)*cov(MVBG),4)
H
```

```
##        [,1]   [,2]   [,3]   [,4]
## [1,] 0.0735 0.5371 0.3321 0.2083
## [2,] 0.5371 4.1995 2.3556 1.6370
## [3,] 0.3321 2.3556 6.1142 3.7813
## [4,] 0.2083 1.6370 3.7813 2.4933
```

```r
E <- (n-1) * cov(RS[RS$rootstock==1,-1])

for (i in 2:k) {
E <- E + (n-1) * cov(RS[RS$rootstock==i,-1])
}
round(E,4)
```

```
##        y1      y2     y3     y4
## y1 0.3200  1.6966 0.5541 0.2171
## y2 1.6966 12.1428 4.3636 2.1102
## y3 0.5541  4.3636 4.2908 2.4817
## y4 0.2171  2.1102 2.4817 1.7225
```

Thus, the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$ are obtained using the following code chunk.

```r
lambdas <- eigen(solve(E) %*% H)$values
round(lambdas,4)
```

```
## [1] 1.8756 0.7904 0.2291 0.0260
```

Therefore, the values of the four test statistics are:

```r
Wilk <- det(E)/det(E+H)
Roy <- lambdas[1]/(1+lambdas[1])
Pillai <- sum(lambdas / (1+lambdas))
LH <- sum(lambdas)

rbind(Wilk,Roy, Pillai,LH)
```

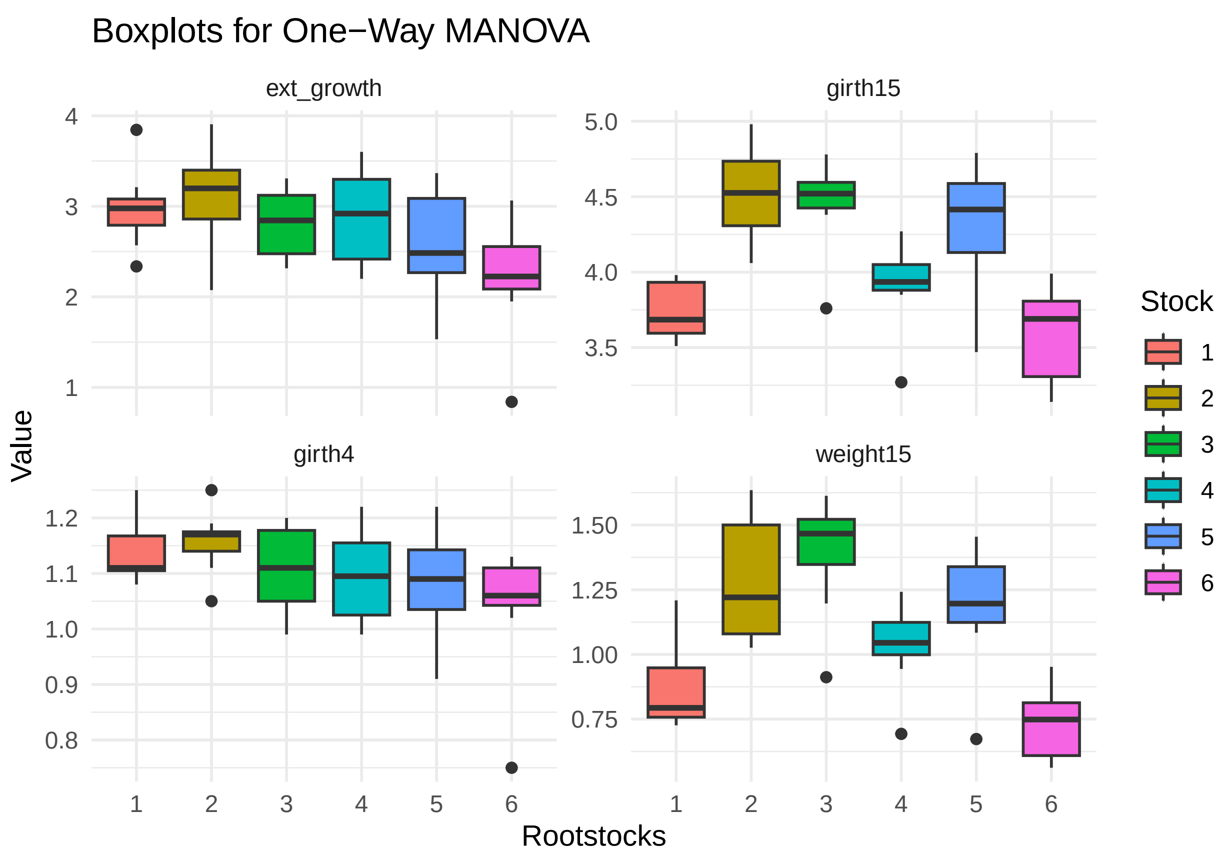
```
##             [,1]
## Wilk   0.1540130
## Roy    0.6522514
## Pillai 1.3054921
## LH     2.9212207
```

Fitting One-way MANOVA in R

Visualizations

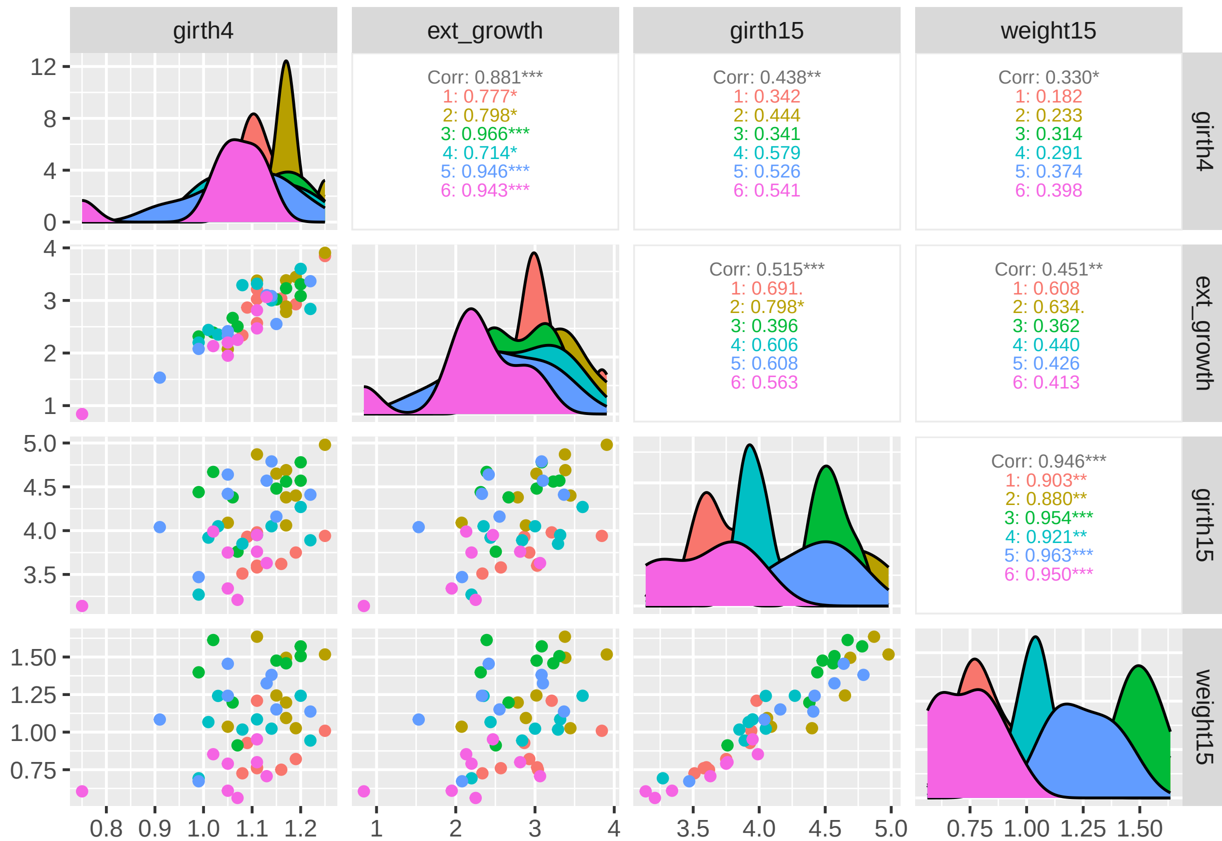
```
RS_long <- pivot_longer(RS1,
                        cols = -c(rootstock,Stock),
                        names_to = "Variable",
                        values_to = "Value")
```

```
ggplot(RS_long, aes(x = Stock, y = Value, fill = Stock)) +
  geom_boxplot() +
  facet_wrap(Variable ~ ., scales = "free_y", ncol = 2) +
  labs(title = "Boxplots for One-Way MANOVA",
       x = "Rootstocks",
       y = "Value") +
  theme_minimal()
```




```
library("GGally")
RS1 |>
```

```
select(girth4:weight15) |>
ggpairs(ggplot2::aes(color=RS1$Stock),
        upper = list(continuous = wrap("cor", size = 2.5)))
```



Generating table of means

```
RS1 |>
  select(-rootstock) %>%
  summarise(
    .by = Stock,
    across(.cols = where(is.numeric),
           .fns = mean))
```

```
## # A tibble: 6 x 5
##   Stock girth4 ext_growth girth15 weight15
##   <fct>  <dbl>      <dbl>   <dbl>    <dbl>
## 1 1       1.14       2.98    3.74    0.871
## 2 2       1.16       3.11    4.52    1.28
```

```
## 3 3        1.11        2.82     4.46     1.39
## 4 4        1.10        2.88     3.91     1.04
## 5 5        1.08        2.56     4.31     1.18
## 6 6        1.04        2.21     3.60     0.735
```

Fitting the MANOVA model

```r
RS_manova <- manova(cbind(girth4,ext_growth, girth15,weight15) ~ Stock,
                  data = RS1)

summary(RS_manova) #default test statistic is Pillai
```

```
##           Df Pillai approx F num Df den Df    Pr(>F)
## Stock      5 1.3055   4.0697     20    168 1.983e-07 ***
## Residuals 42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(RS_manova, test ="Wilks")
```

```
##           Df   Wilks approx F num Df den Df    Pr(>F)
## Stock      5 0.15401   4.9369     20  130.3 7.714e-09 ***
## Residuals 42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(RS_manova, test ="Hotelling-Lawley")
```

```
##           Df Hotelling-Lawley approx F num Df den Df    Pr(>F)
## Stock      5           2.9214   5.4776     20    150 2.568e-10 ***
## Residuals 42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(RS_manova, test ="Roy")
```

```
##           Df    Roy approx F num Df den Df    Pr(>F)
## Stock      5 1.8757   15.756      5     42 1.002e-08 ***
## Residuals 42
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Recovering the H and E matrices

```r
(H_mat <- summary(RS_manova)$SS$Stock)
```

```
##                girth4 ext_growth   girth15 weight15
## girth4     0.07356042  0.5373852 0.3322646 0.208470
## ext_growth 0.53738521  4.1996619 2.3553885 1.637108
## girth15    0.33226458  2.3553885 6.1139354 3.781044
## weight15   0.20847000  1.6371084 3.7810437 2.493091
```

```r
(E_mat <- summary(RS_manova)$SS$Residuals)
```

```
##               girth4 ext_growth   girth15 weight15
## girth4     0.3199875   1.696564 0.5540875 0.217140
## ext_growth 1.6965637  12.142790 4.3636125 2.110214
## girth15    0.5540875   4.363612 4.2908125 2.481656
## weight15   0.2171400   2.110214 2.4816562 1.722525
```

Testing the Assumptions

Assumption: Distribution of residuals are multivariate normal

```r
library(MVN)
RS_resid <- RS_manova$residuals

mvn(data = RS_resid,
    mvn_test = "hz",
    univariate_test = "SW",
    descriptives = FALSE) %>%
  summary()
```

```
##

## -- Multivariate Normality Test Results -----------------------------------------

##            Test Statistic p.value     Method      MVN
## 1 Henze-Zirkler     0.907    0.09 asymptotic   Normal

##
```

```
## -- Univariate Normality Test Results ------------------------------------------

##           Test   Variable Statistic p.value    Normality
## 1 Shapiro-Wilk     girth4     0.952   0.047  Not normal
## 2 Shapiro-Wilk ext_growth     0.978   0.497      Normal
## 3 Shapiro-Wilk    girth15     0.950   0.039  Not normal
## 4 Shapiro-Wilk   weight15     0.973   0.334      Normal
```

Assumption: Distributions of residuals have equal covariance matrices

```r
library(rstatix)
```

```
##
## Attaching package: 'rstatix'

## The following object is masked from 'package:stats':
##
##     filter
```

```r
box_m(data = RS1[,2:5],
      group = RS1$Stock)
```

```
## # A tibble: 1 x 4
##   statistic p.value parameter method
##       <dbl>   <dbl>     <dbl> <chr>
## 1      44.0   0.711        50 Box's M-test for Homogeneity of Covariance Matric~
```

Supposing we reject the null hypothesis that the covariances matrices are all equal, then we need to test for homogeneity of variance for each dependent variable.

```r
RS_long |>
  group_by(Variable) |>
  levene_test(Value ~ Stock)
```

```
## # A tibble: 4 x 5
##   Variable     df1   df2 statistic     p
##   <chr>      <int> <int>     <dbl> <dbl>
## 1 ext_growth     5    42     0.287 0.918
## 2 girth15        5    42     0.623 0.683
## 3 girth4         5    42     0.790 0.563
## 4 weight15       5    42     0.422 0.830
```

Post Hoc Analysis 1: Finding the differences

If we reject the null hypothesis, all we are claiming is that at least one group mean is different for at least one variable being tested.

Since we have 5 variables, we are essentially conducting 5 ANOVAs at once. Let's perform each ANOVA individually!

But how can we do it quickly? By using *anova_test()* function in the **rstatix** package. To do this we need the "long" format of the data.

```
library(tidyverse)
library(rstatix)
RS_long %>%
  group_by(Variable)  %>%
  anova_test(Value ~ Stock)
```

```
## # A tibble: 4 x 8
##   Variable   Effect   DFn   DFd     F           p `p<.05`   ges
## * <chr>      <chr>  <dbl> <dbl> <dbl>       <dbl> <chr>   <dbl>
## 1 ext_growth Stock      5    42  2.90 0.024       "*"     0.257
## 2 girth15    Stock      5    42 12.0  0.000000311 "*"     0.588
## 3 girth4     Stock      5    42  1.93 0.109       ""      0.187
## 4 weight15   Stock      5    42 12.2  0.000000259 "*"     0.591
```

If we had failed the normality test for each dependent variable, we could use a nonparametric test: Kruskal test.

```
RS_long %>%
  group_by(Variable) %>%
  kruskal_test(Value ~ Stock)
```

```
## # A tibble: 4 x 7
##   Variable   .y.       n statistic    df         p method
## * <chr>      <chr> <int>     <dbl> <int>     <dbl> <chr>
## 1 ext_growth Value    48     10.4      5 0.0659    Kruskal-Wallis
## 2 girth15    Value    48     28.8      5 0.0000257 Kruskal-Wallis
## 3 girth4     Value    48      8.24     5 0.144     Kruskal-Wallis
## 4 weight15   Value    48     27.5      5 0.0000458 Kruskal-Wallis
```

Post Hoc 2: Comparing Significant Variables Across Groups

After we determine which variables are different across groups, we want to find which groups are significantly different from one another.

We can use Tukey's Honest Significant Differences (HSD) to determine where the differences occur.

Similar to using *anova_test()*, we need the data to be in the "long" format to use *tukey_hsd()* in the **rstatix** package.

```
RS_long %>%
  filter(Variable %in% c("ext_growth", "girth15", "weight15"))  %>%
  group_by(Variable) %>%
  tukey_hsd(Value ~ Stock)  %>%
  filter(p.adj <0.05)
```

```
## # A tibble: 17 x 10
##    Variable   term  group1 group2 null.value estimate conf.low conf.high   p.adj
##    <chr>      <chr> <chr>  <chr>       <dbl>    <dbl>    <dbl>     <dbl>   <dbl>
##  1 ext_growth Stock 2      6               0   -0.895 -1.70     -0.0919  2.11e-2
##  2 girth15    Stock 1      2               0    0.776  0.299     1.25    2.32e-4
##  3 girth15    Stock 1      3               0    0.716  0.239     1.19    7.53e-4
##  4 girth15    Stock 1      5               0    0.574  0.0967    1.05    1.04e-2
##  5 girth15    Stock 2      4               0   -0.609 -1.09     -0.132   5.61e-3
##  6 girth15    Stock 2      6               0   -0.919 -1.40     -0.442   1.3 e-5
##  7 girth15    Stock 3      4               0   -0.549 -1.03     -0.0717  1.59e-2
##  8 girth15    Stock 3      6               0   -0.859 -1.34     -0.382   4.43e-5
##  9 girth15    Stock 5      6               0   -0.716 -1.19     -0.239   7.53e-4
## 10 weight15   Stock 1      2               0    0.409  0.107     0.712   2.84e-3
## 11 weight15   Stock 1      3               0    0.520  0.218     0.823   9.48e-5
## 12 weight15   Stock 1      5               0    0.310  0.00760   0.612   4.16e-2
## 13 weight15   Stock 2      6               0   -0.546 -0.848    -0.243   4.23e-5
## 14 weight15   Stock 3      4               0   -0.352 -0.655    -0.0501  1.4 e-2
## 15 weight15   Stock 3      6               0   -0.656 -0.959    -0.354   1.17e-6
## 16 weight15   Stock 4      6               0   -0.304 -0.606    -0.00172 4.8 e-2
## 17 weight15   Stock 5      6               0   -0.446 -0.748    -0.144   9.55e-4
## # i 1 more variable: p.adj.signif <chr>
```

Or, when using the Kruskal-Wallis test the most common (post hoc) pairwise comparison test is dunn's test. We can do this with the *holm* correction or adjustment to control family-wise error rate. It guarantees that the probability of making at least one Type I error across the family of tests is less than or equal to (α).

```
RS_long %>%
  filter(Variable %in% c("ext_growth", "girth15", "weight15"))  %>%
  group_by(Variable) %>%
  dunn_test(Value ~ Stock,
            p.adjust.method = "holm") %>%
  filter(p.adj <0.05)
```

```
## # A tibble: 10 x 10
##    Variable .y.   group1 group2    n1    n2 statistic          p    p.adj
##    <chr>    <chr> <chr>  <chr>  <int> <int>     <dbl>      <dbl>    <dbl>
##  1 girth15  Value 1      2          8     8      3.57 0.000354  0.00496
##  2 girth15  Value 1      3          8     8      3.27 0.00108   0.0130
##  3 girth15  Value 2      6          8     8     -3.84 0.000123  0.00185
##  4 girth15  Value 3      6          8     8     -3.54 0.000406  0.00528
##  5 girth15  Value 5      6          8     8     -3.00 0.00270   0.0297
##  6 weight15 Value 1      2          8     8      2.87 0.00415   0.0457
##  7 weight15 Value 1      3          8     8      3.40 0.000669  0.00870
##  8 weight15 Value 2      6          8     8     -3.71 0.000211  0.00295
##  9 weight15 Value 3      6          8     8     -4.24 0.0000222 0.000333
## 10 weight15 Value 5      6          8     8     -3.01 0.00262   0.0314
## # i 1 more variable: p.adj.signif <chr>
```

Games-Howell test

If we failed the equal variance assumption, we can use a more robust test: Games-Howell test

```
RS_long |>
  group_by(Variable) |>
  games_howell_test(Value ~ Stock) %>%
  filter(p.adj<0.05)
```

```
## # A tibble: 15 x 9
##    Variable .y.   group1 group2 estimate conf.low conf.high   p.adj p.adj.signif
##    <chr>    <chr> <chr>  <chr>     <dbl>    <dbl>     <dbl>   <dbl> <chr>
##  1 girth15  Value 1      2         0.776  0.307      1.25   2   e-3 **
##  2 girth15  Value 1      3         0.716  0.285      1.15   1   e-3 ***
##  3 girth15  Value 1      5         0.574  0.00623    1.14   4.7 e-2 *
##  4 girth15  Value 2      4        -0.609 -1.13      -0.0893 1.8 e-2 *
##  5 girth15  Value 2      6        -0.919 -1.47      -0.370  9.01e-4 ***
##  6 girth15  Value 3      4        -0.549 -1.04      -0.0585 2.5 e-2 *
##  7 girth15  Value 3      6        -0.859 -1.38      -0.336  1   e-3 ***
##  8 girth15  Value 5      6        -0.716 -1.34      -0.0937 2.1 e-2 *
##  9 weight15 Value 1      2         0.409  0.0664     0.752  1.6 e-2 *
## 10 weight15 Value 1      3         0.520  0.184      0.856  2   e-3 **
## 11 weight15 Value 2      6        -0.546 -0.875     -0.216  2   e-3 **
## 12 weight15 Value 3      4        -0.352 -0.692     -0.0127 4   e-2 *
## 13 weight15 Value 3      6        -0.656 -0.978     -0.334  2.36e-4 ***
## 14 weight15 Value 4      6        -0.304 -0.564     -0.0442 1.8 e-2 *
## 15 weight15 Value 5      6        -0.446 -0.781     -0.111  8   e-3 **
```